Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution Affiliated to Anna University)

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Experiment 4: Binary Classification using Linear and Kernel-Based Models

Objective

The objective of this experiment is to classify emails as spam or non-spam using Logistic Regression and Support Vector Machine (SVM) models and to analyze the effect of kernel functions and hyperparameter tuning on classification performance.

Dataset

The Spambase dataset contains numerical features extracted from email content along with a binary target variable indicating whether an email is spam or non-spam (ham).

- Kaggle Dataset: Spambase Dataset

3. Preprocessing Steps

3.1 Missing Value Check

The dataset was examined for missing values. No missing values were found and hence no imputation was required.

3.2 Feature Standardization

All features were standardized using `StandardScaler` to ensure zero mean and unit variance. Feature scaling is essential for distance-based and margin-based classifiers such as Logistic Regression and SVM.

3.3 Train–Test Split

The dataset was divided into training and testing sets in an 80:20 ratio. The training set was used for learning model parameters, while the testing set was used for evaluation.

4. Implementation Details

4.1 Logistic Regression

Logistic Regression was implemented as a linear classification model. Both $L1$ and $L2$ regularization techniques were evaluated. The inverse regularization parameter C was tuned to obtain optimal model performance.

4.2 Support Vector Machine

Support Vector Machines were trained using the following kernel functions:

- Linear
- Polynomial
- Radial Basis Function (RBF)
- Sigmoid

Hyperparameters such as C and γ were optimized using grid search.

4.3 Validation Strategy

A 5-fold cross-validation technique was used to ensure reliable performance estimation.

5. Exploratory Data Analysis

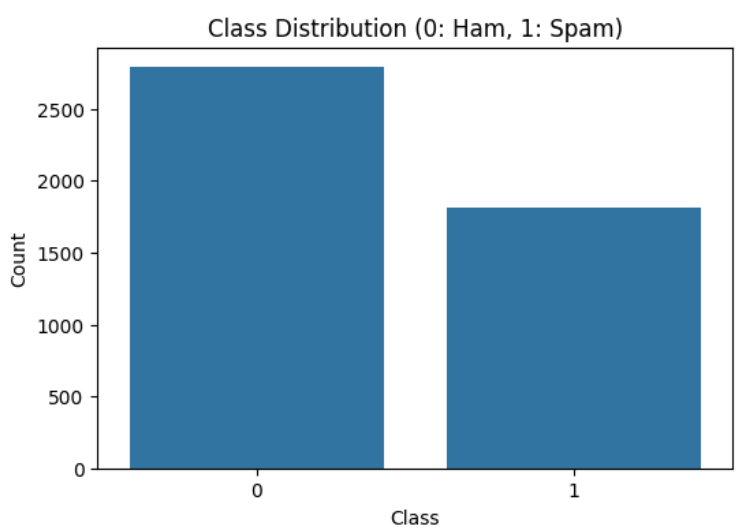

Figure 1: Spam vs Ham Class Distribution

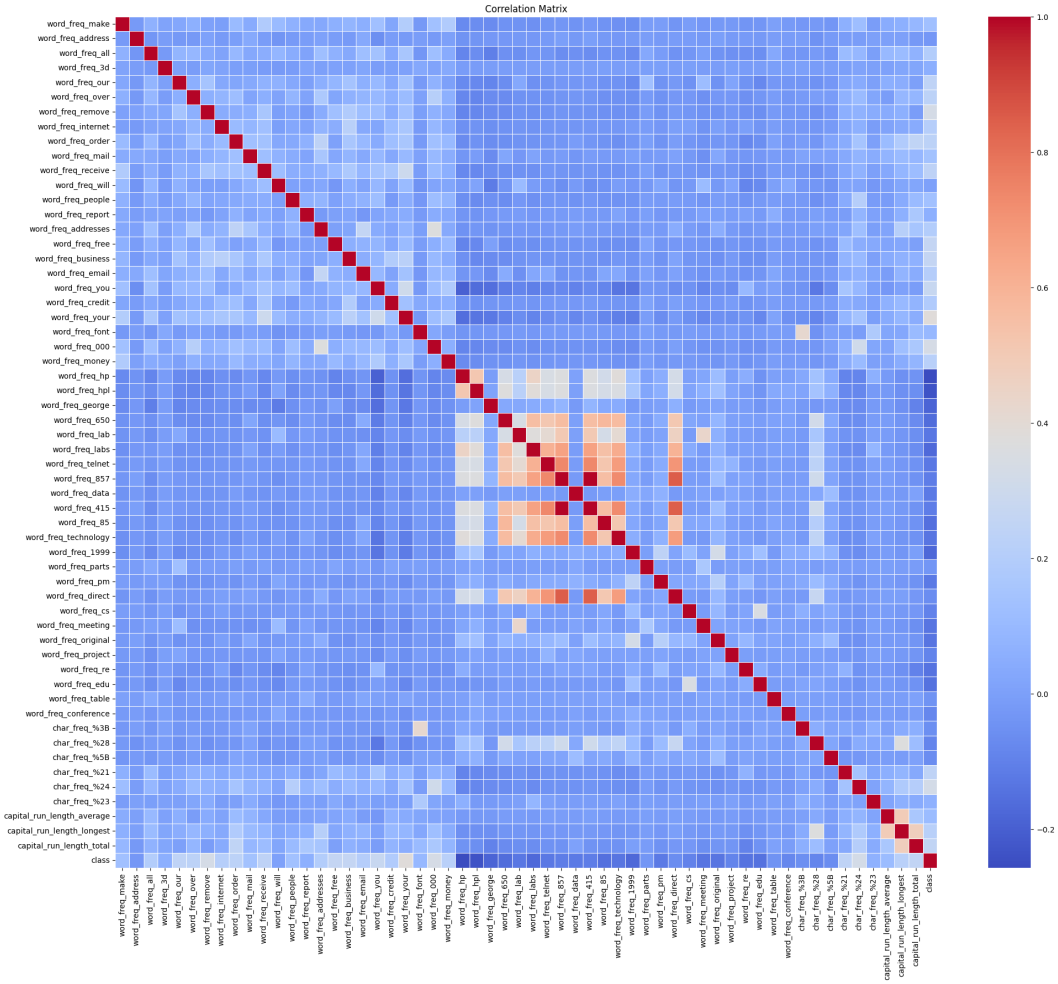

Figure 2: Correlation Heatmap

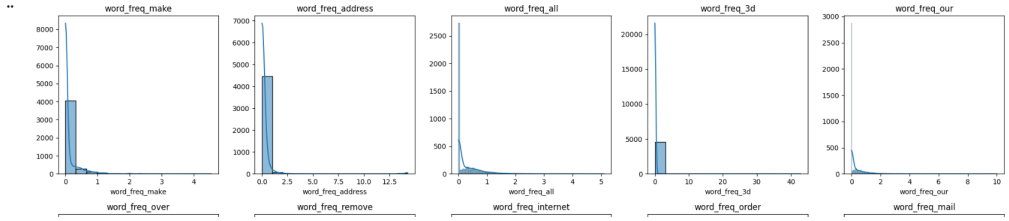

Figure 3: Histogram of Feature Distributions

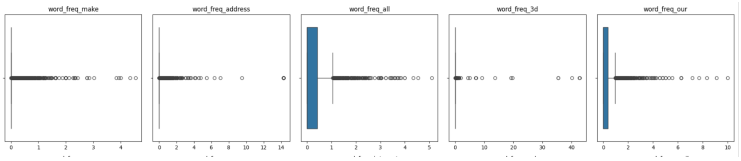

Figure 4: Boxplot for Outlier Detection

6. Results and Performance Analysis

This section presents the experimental results obtained from baseline models, hyperparameter tuning, kernel analysis, and final model evaluation.

6.1 Baseline Logistic Regression Performance

A baseline Logistic Regression model was trained using default hyperparameters to establish an initial performance benchmark.

Metric	Value
Training Time (s)	0.0204
Accuracy	0.9300
Precision	0.9200
Recall	0.9000
F1 Score	0.9100

6.2 Hyperparameter Tuning Results

GridSearchCV with 5-fold cross-validation was used to tune hyperparameters for both Logistic Regression and Support Vector Machine models.

Model	Best Parameters	CV Accuracy
Logistic Regression	C = 1, L1 Regularization (saga)	0.9239
Support Vector Machine	C = 10, Kernel = RBF, Gamma = scale	0.9342

6.3 SVM Kernel-wise Performance Comparison

To analyze the impact of kernel functions, SVM models were trained using four different kernels.

Kernel	Accuracy	F1 Score	Training Time (s)
Linear	0.9164	0.8972	0.1405
Polynomial	0.7796	0.6220	0.2210
RBF	0.9273	0.9055	0.1361
Sigmoid	0.8838	0.8524	0.1403

6.4 Final Model Performance (Best Estimators)

The best-performing models selected through grid search were evaluated on the test dataset.

Logistic Regression (Best Model)

Metric	Value
Training Time (s)	5.4336
Accuracy	0.9294
Precision	0.9209
Recall	0.8981
F1 Score	0.9093

Support Vector Machine (Best Model)

Metric	Value
Training Time (s)	0.1164
Accuracy	0.9229
Precision	0.9195
Recall	0.8815
F1 Score	0.9001

6.5 K-Fold Cross-Validation Results (K = 5)

To evaluate the stability and generalization capability of the models, 5-fold cross-validation was performed.

Fold	Logistic Regression	SVM
Fold 1	0.9197	0.9359
Fold 2	0.9283	0.9283
Fold 3	0.9304	0.9435
Fold 4	0.9413	0.9380
Fold 5	0.8337	0.8522
Average	0.9107	0.9196

6.6 Summary of Results

- Baseline Logistic Regression achieved an accuracy of approximately 93%.
- Hyperparameter tuning improved model stability and generalization.
- The RBF kernel consistently outperformed other SVM kernels.
- Logistic Regression achieved slightly higher final test accuracy than SVM.
- Cross-validation results confirmed consistent performance across folds.

Learning Outcomes

- Learned binary classification techniques.
- Understood regularization and kernel methods.
- Implemented hyperparameter tuning using GridSearchCV.
- Evaluated models using multiple performance metrics.

References

- Scikit-learn – Logistic Regression
- Scikit-learn – Support Vector Machines
- Spambase Dataset